Low Level Design Document

Introduction

This Low Level Design (LLD) document outlines the implementation details for **LoanForest - Loan Default Predictor**, a Streamlit application that predicts loan default risk using a Random Forest model. The app provides an interactive UI for users to input loan features and receive a default risk prediction.

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1. System Components

Component	Description	Key Responsibilities
UI Layer	Streamlit-based web interface	Collect user input, display results
Model Layer	Random Forest model (scikit-learn)	Predict default risk
Data Processing	Input validation & feature transformation	Prepare data for model
Controller	Orchestrates UI, data processing, and prediction	Handle app flow

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2. Class/Interface Overview

Class/Module	Description	Key Methods/Attributes
LoanForestApp	Main app controller	`run()`, `get_user_input()`, `predict()`
LoanDefaultModel	Model wrapper	`load_model()`, `predict(features)`
FeatureSchema	Data structure for input features	Attributes: `age`, `income`, …

Relationships:

- `LoanForestApp` uses `LoanDefaultModel` for predictions and `FeatureSchema` for input handling.

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3. Data Structure Overview

Field Name	Type	Description
age	int	Applicant's age
income	float	Annual income
loan_amount	float	Requested loan amount
term	int	Loan term (months)
credit_score	int	Credit score
…	…	Additional relevant features

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4. Algorithms/Logic

Prediction Flow:

```
def run():
    features = get_user_input()
    if validate(features):
        processed = preprocess(features)
        risk = model.predict(processed)
        display_result(risk)
    else:
        display_error("Invalid input")
```

- **get_user_input()**: Collects feature values via Streamlit sliders/inputs.
- **preprocess()**: Converts raw input to model-ready format.
- **model.predict()**: Returns default risk probability/class.

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5. Error Handling

Scenario	Handling Approach
Invalid user input	Show error message in UI
Model file not found/corrupt	Display critical error, disable predict
Prediction failure	Log error, show generic error message
Data preprocessing error	Show input validation error

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